

Unimolecular Reactions in a Spherically Symmetric Potential. 3. Lifetimes of Collision Complexes

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We describe an analytical procedure for obtaining rate constants, branching ratios, and kinetic energy distributions following the unimolecular dissociation of a species with arbitrary energy and angular momentum. With it the effects of small changes in the assumed potential energy surface can be readily examined. Limitations to the Rice–Gershinowitz hypothesis concerning angular momentum effects are identified. It is further shown how angular momentum can affect a branching ratio and thus, indirectly, a kinetic energy distribution.

Introduction

The reactive scattering of ions from other molecules has both a considerable history¹ and a lively present-day following.² Of especial interest are those reactions which seem to go by way of a collision complex which persists for at least the length of a rotational period. They raise the possibility of using unimolecular reaction theory to predict their outcome. The earliest example of a chemical activation experiment in fact made use of an ion–molecule collision.³

Because of the high angular momentum which a complex can have, a calculation of its statistical evolution has never been straightforward. A convenient recourse has been to assume that the fragments from any such collision separate under the influence of a spherically symmetric potential. Even so, procedures for executing such a calculation can be tedious.^{4,5}

This situation has been ameliorated in two ways. Instead of the usual ion-induced dipole potential, the more realistic form

$$V(R) = -C_4/R^2(R^2 - R_0^2) \quad (1)$$

(with $R \geq R_0$) can be used with no loss of simplicity.⁶ Thus the quantity R_0 can be used to simulate the potential well offered by the postulated complex. By varying its magnitude, one can also obtain a sense of the importance of the remaining deficiencies in the potential. An analogous formulation to describe the dissociation of neutral species has also been given.⁶

An experimentally accessible property which can bear the impress of that potential is the distribution of kinetic energies borne by the separating fragments. Recently it has been shown how that distribution can be calculated analytically.⁷ It is thus now possible to display rapidly the effects of energy, angular momentum, isotopic substitution, and the parameters of the assumed potential on the predicted kinetic energy distribution.

It is desirable to have an equally simple way for predicting lifetimes for a complex. To some extent they are experimentally accessible from the angular distribution of the products. In any case they can indicate whether the idea of an ergodic complex is at all tenable, and thus whether any agreement (or lack of it)

between theory and experiment can be considered meaningful. In this paper we describe such a procedure. It is in fact nothing more than a conflation of earlier results. Our procedure will be tested by showing that it goes over to well-known relations in a number of limiting cases. Examples of how it can be more generally useful will then be indicated.

Rate Constants

We consider a molecule containing energy $E + E_0$ as measured from its zero point, where E_0 is the thermodynamic threshold for a given reaction channel. The quasiequilibrium rate constant for reaction via that channel can be written as

$$k(E, J) = W(E, J)/h\rho(E + E_0, J) \quad (2)$$

where $\rho(E + E_0, J)$ is the density of states of the parent species, with an angular momentum indicated by the number J . We will be concerned only with the function $W(E, J)$. The extreme case of a highly-attractive potential surface, for example with the potential parameters C_4 and R_0 arbitrarily large, is known as the phase-space limit. When, in addition, the angular momentum is arbitrarily small, this function is given by⁸

$$W_{ps}(J \sim 0) = (2J + 1)^2 \int_0^E \rho_v(x)(E - x)^s dx/\beta \quad (3)$$

where $\rho_v(x)$ is the density of vibrational levels of the products, where s equals $(r - 1)/2$ where r is the number of their rotational degrees of freedom, and where β is a tabulated function⁸ of their rotational constants, referred to below as B_1 and B_2 .

One can show easily that any relaxation of the criteria which define this limiting case will lead to a smaller rate constant than the one implied by eq 3. We therefore generalize it by multiplying it by an integral over the known function⁷ (see also below) for the kinetic energy distribution, generally valid for spherically symmetric potentials, then normalized so that the integral is unity in the appropriate limit. Accordingly we write

$$W(E, J) = W_{ps}(J \sim 0) N \int f(\epsilon, J) \exp(-\epsilon/k_B T^*) d\epsilon/k_B T^* \quad (4)$$

where the temperature is defined by

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$$E + E_0 = \bar{E}_p(T^\ddagger, J) + \Delta E_a(T^\ddagger, J) - k_B T^\ddagger \quad (5)$$

where $\bar{E}_p$ and ΔE_a are respectively the canonical energy of the parent and the Arrhenius activation energy for the reaction and where the normalization factor is given by

$$N = (k_B T^\ddagger / B J^2)^{1/2} \Gamma(3/2)/2 \quad (6)$$

with $B = B_1 B_2 / (B_1 + B_2)$. We then denote by $L_m(\epsilon)$ the maximum value of the orbiting angular momentum quantum number which permits passage of a collision pair with kinetic energy ϵ . On a surface described by eq 1, for example, it is given by⁶

$$L_m^2(\epsilon) = (2\mu/\hbar^2)[R_0^2\epsilon + (4C_4\epsilon)^{1/2}] \quad (7)$$

where μ is the reduced mass. We recall finally that the function $f(\epsilon, J)$ is given,⁷ when $L_m(\epsilon) > J$, by

$$f(\epsilon, J) = 2 \operatorname{erf}(B J^2 / k_B T^\ddagger)^{1/2} - \operatorname{erf}(B(L + J)^2 / k_B T^\ddagger)^{1/2} + \operatorname{erf}(B(L - J)^2 / k_B T^\ddagger)^{1/2} \quad (8a)$$

and, when $L_m(\epsilon) < J$, by

$$f(\epsilon, J) = 2 \operatorname{erf}(B J^2 / k_B T^\ddagger)^{1/2} - \operatorname{erf}(B(L + J)^2 / k_B T^\ddagger)^{1/2} - \operatorname{erf}(B(L - J)^2 / k_B T^\ddagger)^{1/2} \quad (8b)$$

It is clear how these relations should be used. For a given energy and angular momentum, one needs to evaluate eq 8, and then the integral over it, at two different (and more or less arbitrarily chosen) temperatures to obtain a J -dependent contribution to the activation energy. It must be added to the J -independent contribution arising from the phase-space rate constant, which is then entered into eq 5 to obtain the temperature at which eq 4 should finally be evaluated. Small corrections, as suggested in the Appendix of ref 7, may then be made.

Before considering such applications, however, we examine whether it leads to what can be otherwise obtained in some well-defined limits. The normalization factor in eq 6 has already ensured that we recover the small- J phase space result, such as occurs when either R_0 or C_4 in eq 7 becomes arbitrarily large. Consider then a potential surface so attractive that we are still in the phase space limit, but now with the angular momentum arbitrarily large. The right-hand side of eq 8 is again determined only by the first of its error function components, which is now unity. Upon recalling a steepest-descent theorem (in the Appendix of ref 5), one then finds agreement with the phase-space results worked out by Troe⁹ for the high- J limit.

Olzmann and Troe¹⁰ have examined the phase-space limit for intermediate values of J by way of a large number of ad hoc calculations. They report that these can be represented by a universal curve given in their Figure 2 and which they fit with a complicated expression using the notation $y = f(x)$. The present results imply that their data can be represented by the much simpler expression

$$y = \operatorname{erf}(x\Gamma(3/2)) \quad (9)$$

Inspection of their figure shows this to be the case.

Consider finally the opposite limit from that of phase-space theory, e.g. an only weakly attractive potential surface where the potential parameters C_4 and R_0 are now arbitrarily small. This important limit is physically realized in the evaporation of a small species from a large body, and in thermionic emission,

not because the potential parameters are small but because the ratio $BL_{\max}^2/k_B T$ can be much less than unity in those circumstances. We obtain then

$$f(\epsilon) = 2J \exp(-B J^2 / k_B T^\ddagger) (BL_{\max}^2 / k_B T^\ddagger) \quad (10)$$

This is just the previously identified thermodynamic limit;¹¹ the kinetic energy distribution and (after averaging over a Boltzmann distribution of energies) the rate constant will be those which one will observe in an unconstrained heat bath.

Activation Energies

It was mentioned above that the Arrhenius activation energy required in eq 5 could always be obtained by evaluating eq 4 at two different temperatures. We first examine whether a simpler procedure exists. It was postulated recently¹² that the angular momentum dependence of a unimolecular rate constant could be represented by the expression

$$\delta \ln k(E, J) / \delta J^2 = B^\circ / k_B T - B^\ddagger / k_B T^\ddagger \quad (11)$$

where B° and $B^\ddagger$ are the rotational constants of the parent and its transition state, respectively, and where T is defined by the expression

$$E + E_0 = \bar{E}_p(T) - k_B T \quad (12)$$

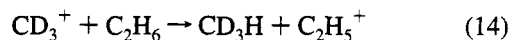
That relation was obtained by assuming, following Rice and Gershinowitz,¹³ that the role of angular momentum is to increase the Arrhenius activation energy by an amount $(B^\ddagger - B^\circ)J^2$. Further, in the special case of a spherically symmetric potential, when the transition state is diffuse and hence $B^\ddagger$ seemingly poorly defined, a simple relation for it has been given.¹²

This possibility can be tested as follows. The density of states of the parent species in eq 2 will contain a factor $(2J + 1)^2$ exactly canceling that factor in eq 3. The decrease in the density of the parent's vibrational states with increasing J is meanwhile given by the first term on the right-hand side of eq 11. The second term then implies that the integral over $f(\epsilon)$ in eq 4, together with the normalization factor N , can be written as

$$[N \operatorname{Int}]_J = [N \operatorname{Int}]_0 \exp(-B^\ddagger J^2 / k_B T^\ddagger) \quad (13)$$

Note that the thermodynamic limit, described in eq 10, conforms to that relation. Thus the Rice–Gershinowitz assumption, which might have been expected to apply only to “tight”, localized transition states, pertains also to at least one limit of those derived from spherically-symmetric potentials. Let us see if it is more general.

We consider the reaction



and assume 1×10^{-4} and 2.9×10^{-4} eV for B_1 and B_2 , the rotational constants of its product fragments, and a value of 1.9×10^{-31} eV cm⁴ for C_4 , the associated Langevin parameter. Two values of R_0 , the infinite sink radius, zero and 3×10^{-8} cm, were then chosen. Figure 1 illustrates calculations of these integrals over the kinetic energy distribution, for several values of angular momentum but always using the same arbitrarily chosen value for $k_B T^\ddagger$ of 0.15 eV. Included also are the predictions of eq 13, using the prescription

$$(B^\ddagger/B)^{3/2} = \int \exp(-\epsilon/k_B T^\ddagger) [1 - \exp(-BL_{\max}^2/k_B T^\ddagger)] d\epsilon / \int \exp(-\epsilon/k_B T^\ddagger) (BL_{\max}^2/k_B T^\ddagger) d\epsilon \quad (15)$$

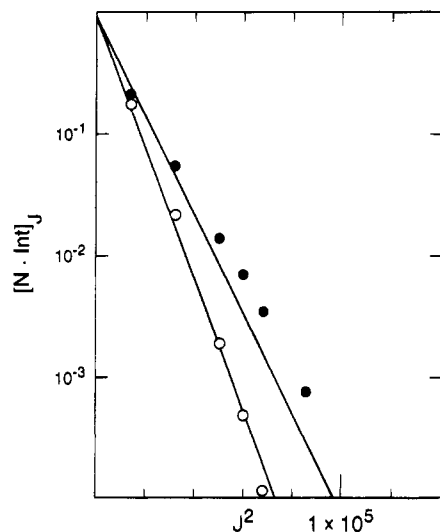


Figure 1. Effect of angular momentum on the normalized integral over kinetic energy distributions: (open and closed circles) R_0 , the infinite sink radius, equal to zero and 3 Å respectively; (solid lines) the predictions of eq 15.

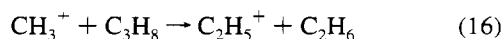
which follows if the Rice–Gershinowitz assumption is strictly correct.¹²

It will be seen that when R_0 is zero, the angular momentum dependence is in good agreement with expectations. The potential surface is evidently sufficiently weak that while it is not yet in the thermodynamic limit, the functional dependence predicted for that limit is observed. But, with an increase in the magnitude of R_0 , one can move quickly away from that happy state. Serious departures from simple expectations become apparent.

Because of these departures, it is clearly preferable to obtain Arrhenius activation energies *in situ* than from the Rice–Gershinowitz ansatz. Nevertheless, for qualitative arguments, eq 11 should continue to be useful for discussing the effect of angular momentum on a rate constant. That effect will be seen to be a trade-off between two opposing tendencies. The one which dominates will depend upon such matters as the details of the potential surface and the distance from the reaction threshold.

Branching Ratios and Kinetic Energy Distributions

Equation 9 contains a relation for the spectrum of kinetic energies borne by the products of a reaction. As a preliminary matter, we see what effect small changes in the assumed potential surface should have on any such distribution. The reaction



will be used as an example. Plausible values for the moments of inertia and vibrational frequencies of the products, and for C_4 , the ion-induced dipole interaction coefficient, can be readily settled on. A collision complex with 4.0-eV energy greater than the reaction threshold will be assumed. Initially the distribution of angular momenta will be assumed to be given by¹⁴

$$p(J^2) = \text{constant} \quad (17)$$

with J in the interval $0 \leq J \leq 345$, as might seem appropriate for complexes formed with 1.85-eV center-of-mass energy. Figure 2 illustrates the predicted kinetic energy distributions

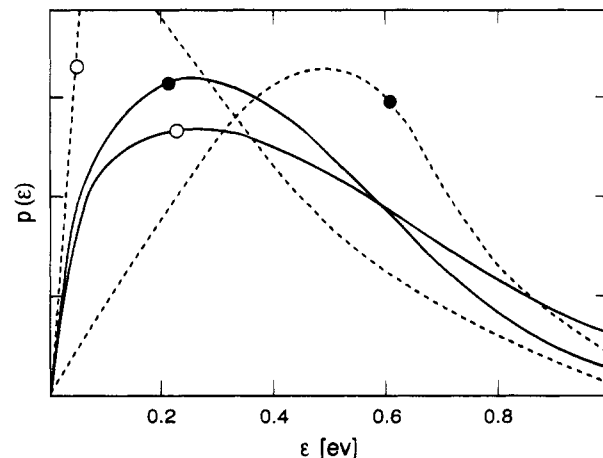
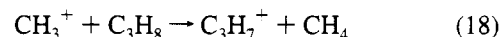


Figure 2. Calculated kinetic energy distributions for reaction 16 with (dashed lines) and without (solid lines) competition from reaction 18: (open and closed circles) R_0 equal to zero and 3 Å, respectively.

for values of R_0 , the infinite well radius, equal to zero and 3 Å. They will be seen to be very nearly the same. We may suspect that remaining ambiguities in the potential surface will be of little effect.

Actually reaction¹⁶ should be parasitic to the much more exothermic process



The branching ratio, while always small, is expected to be strongly dependent on angular momentum such that eq 17 will surely be inappropriate to describe the origins of whatever C_2H_5^+ is observed. When this competition is considered, using the more complete formulation in eq 4, the predicted kinetic energy distributions are quite different, as also illustrated in the figure. When R_0 is zero the predicted distribution now lies predominantly at smaller kinetic energies, whereas when it is 3 Å, the distribution is pushed to higher energy.

These shifts can be understood by considering the second term on the right side of eq 11. The reaction with the higher threshold will, following eq 5, have a smaller associated value of $k_B T^\ddagger$. All other things being equal, angular momentum will thus mitigate against that reaction more severely than against a competing channel. Those products which are observed will have tended to come from complexes with low angular momentum, and the kinetic energy distribution should reflect that tendency. As seen in Figure 1, however, making the potential surface more attractive lowers the value of $B^\ddagger$ and moderates the impact of the angular momentum. One can reach the point where the kinetic energy distribution instead predominately originates with the high- J collision complexes. Thus for minor reaction channels, a predicted distribution can be quite sensitive to the assumed parameters, and comparisons with experiment will be accordingly less meaningful.

Summary

The procedure described above is capable of rapidly generating rate constants, branching ratios, and kinetic energy distributions for the unimolecular dissociation of molecules with arbitrary energy and angular momentum. It is easily programmed so that the effects of changes in either these or any other of the necessary input parameters are readily examined. Copies of such programs are available from the J. Heyrovský Institute.

This dexterity has afforded the possibility of examining the Rice–Gershinowitz hypothesis as it pertains to “loose” transition states. We have seen how, even in this least promising case, there exists a limit where the hypothesis is exactly correct. Outside that limit, deviations from its predictions occur, but for qualitative purposes it should still be quite useful. In particular the large effect which angular momentum can have on the kinetic energy distribution of the products from a secondary reaction channel can be easily comprehended.

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